An Interpretable Machine Learning Model for Predicting Muscle Loss Risk in Oral Cancer Patients Post-Radiotherapy Using Clinical and Treatment-Related Symptom Features

Ning-Hsiang Weng [a], Jie Lee [b], Kun-Pin Wu [a]

[a] Institute of Biomedical Informatics, National Yang Ming Chiao Tung University, Taipei, Taiwan

[b] Department of Radiation Oncology, MacKay Memorial Hospital, Taipei, Taiwan

Introduction

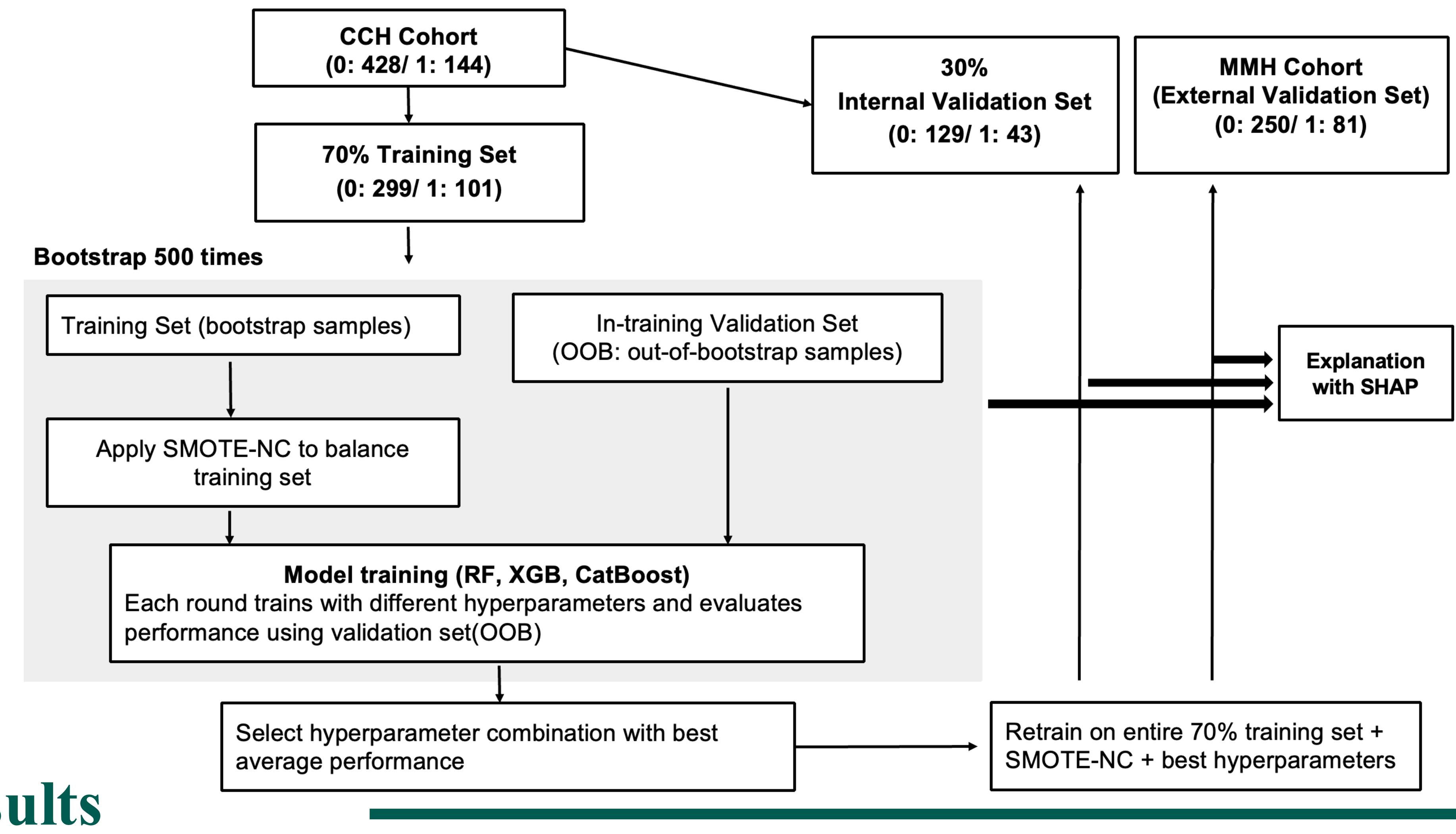
- Oral cavity cancer remains a major public health concern. According to the 2023 Taiwan Cancer Registry report, it ranks fourth in both incidence and mortality among the top ten cancers in Taiwanese males.
- Although radiotherapy improves tumor control, treatment-related adverse effects such as oral mucositis, dysphagia, pain, anorexia, nausea, and fatigue can impair oral intake and nutritional status. Anorexia and systemic muscle wasting are central manifestations of cancer cachexia, while dysphagia further restricts nutritional intake and may aggravate progressive muscle loss.[1]
- Significant muscle loss is associated with reduced treatment tolerance, increased complications, poorer quality of life, and diminished survival.
- **Aim:** To develop an interpretable on-treatment machine learning model integrating baseline clinical characteristics and treatment-related adverse effects to identify oral cancer patients at high risk of post-radiotherapy muscle loss.

Methods

Population: This retrospective multicenter study included 903 oral cancer patients treated between 2010 and 2020. The Changhua Christian Hospital cohort was divided into training (n = 400) and internal validation (n = 172) sets, while the MacKay Memorial Hospital cohort served as an external validation set (n = 331).

Muscle Loss Definition: Significant muscle loss was defined as a ≥ 4.2% decrease in L3-equivalent Skeletal Muscle Index (SMI) converted from C3-level CT images, a threshold associated with significantly increased all-cause mortality[2].

Model Features: Eighteen predictors were included: age, sex, ECOG performance status, tumor site, pathological stage, chemotherapy, smoking, pre-treatment BMI, BMI change, pre-treatment SMI, dermatitis, mucositis, xerostomia, dysphagia, pain, anorexia, nausea, and fatigue. Treatment-related adverse effects were obtained from weekly clinical records, with the highest recorded grade used for analysis.



Results

Training Set

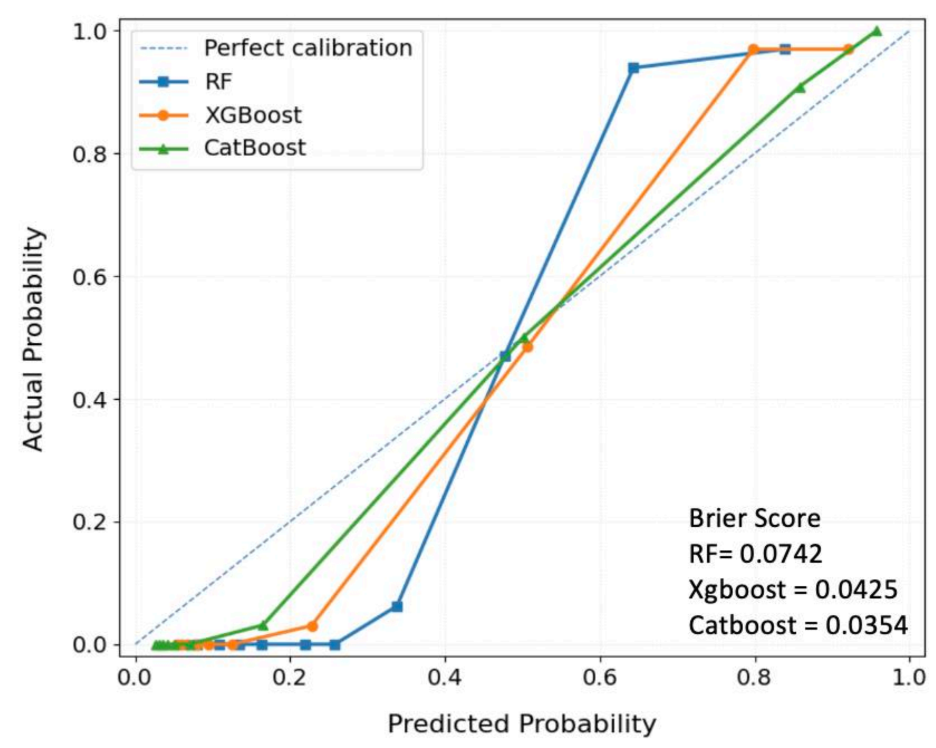
Model	F1 score (95% CI)	ROC-AUC (95% CI)	PR-AUC (95% CI)	Sensitivity (95% CI)	Specificity (95% CI)	Precision (95% CI)	Accuracy (95% CI)
RF	0.785 [b]	0. 942 [a]	0.864 [b]	0.754 [b]	0.946 [a]	0.826 [a]	0.897
	(0.694, 0.86)	(0.914, 0.969)	(0.792, 0.922)	(0.597, 0.893)	(0.901, 0.979)	(0.723, 0.914)	(0.86, 0.929)
XGBoost	0. 796 [a]	0.939 [b]	0.868 [a]	0.796 [a]	0.932 [b]	0.801 [b]	0.898
	(0.716, 0.857)	(0.908, 0.969)	(0.802, 0.925)	(0.642, 0.917)	(0.88, 0.972)	(0.69, 0.895)	(0.865, 0.928)
CatBoost	0. 794 [a]	0.942 [a]	0.869 [a]	0.787 [b]	0.935 [b]	0.806 [b]	0.898
	(0.717, 0.86)	(0.911, 0.971)	(0.797, 0.926)	(0.639, 0.907)	(0.889, 0.972)	(0.708, 0.897)	(0.866, 0.931)

Differences between models within the same column were assessed using Kruskal-Wallis and Dunn's post-hoc tests. Values with different superscript letters are significantly different ($p < 0.05$).

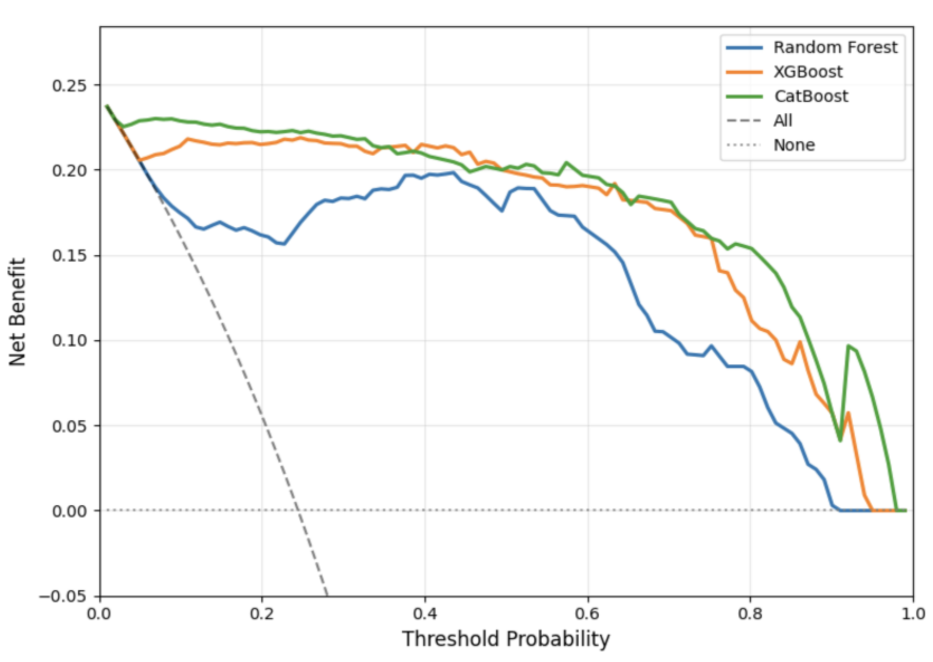
Internal & External Validation Set

Model	F1 score	ROC-AUC	PR-AUC	Sensitivity	Specificity	Precision	Accuracy
Internal Validation							
RF	0.878	0.968	0.929	0.837	0.977	0.923	0.942
XGBoost	0.835	0.964	0.927	0.767	0.977	0.917	0.924
CatBoost	0.84	0.957	0.823	0.791	0.969	0.895	0.924
External Validation							
RF	0.866	0.986	0.957	0.839	0.968	0.895	0.937
XGBoost	0.909	0.99	0.964	0.926	0.964	0.893	0.955
CatBoost	0.909	0.99	0.966	0.926	0.964	0.893	0.955

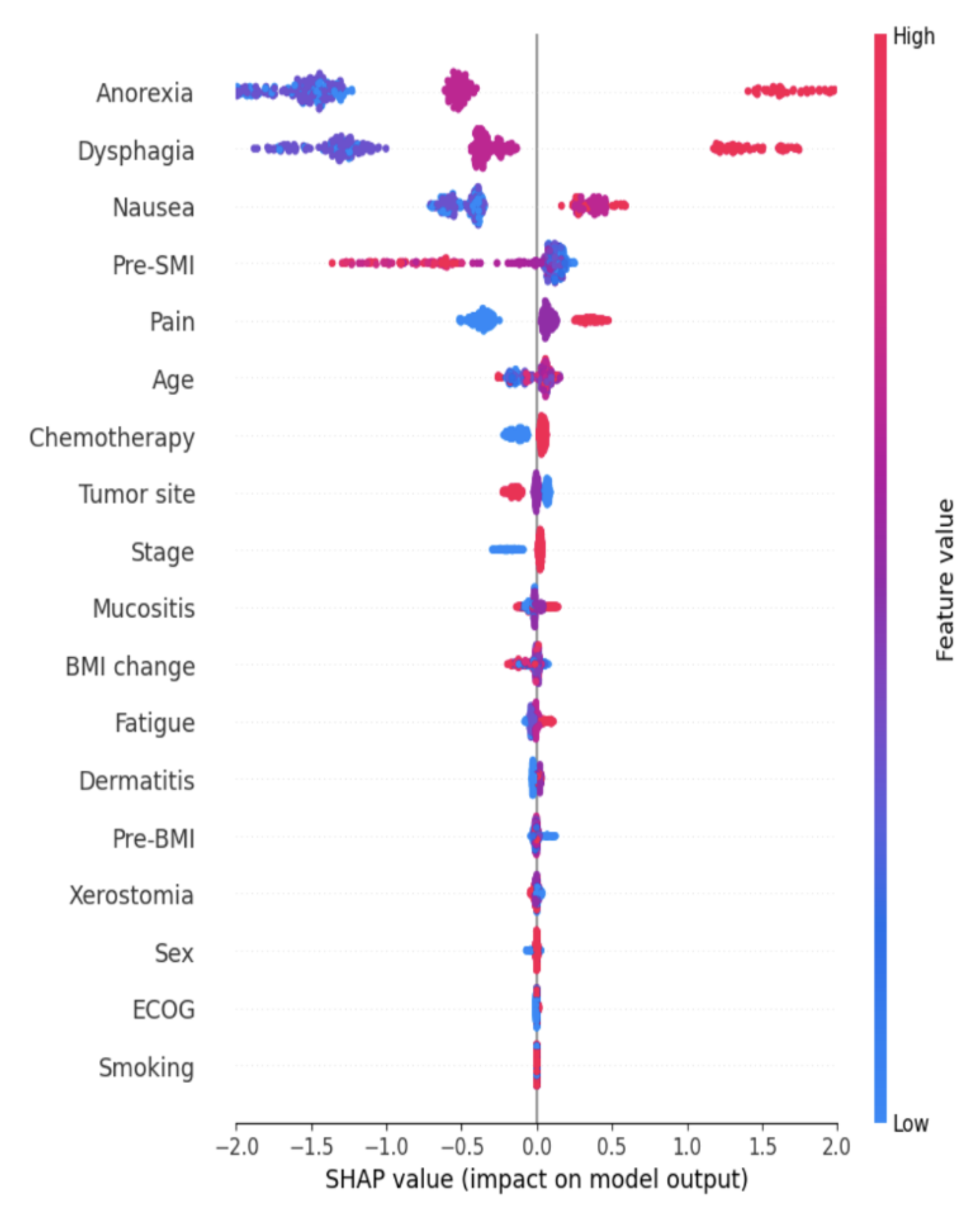
No significant differences in ROC-AUC were observed among the three models (DeLong's test).



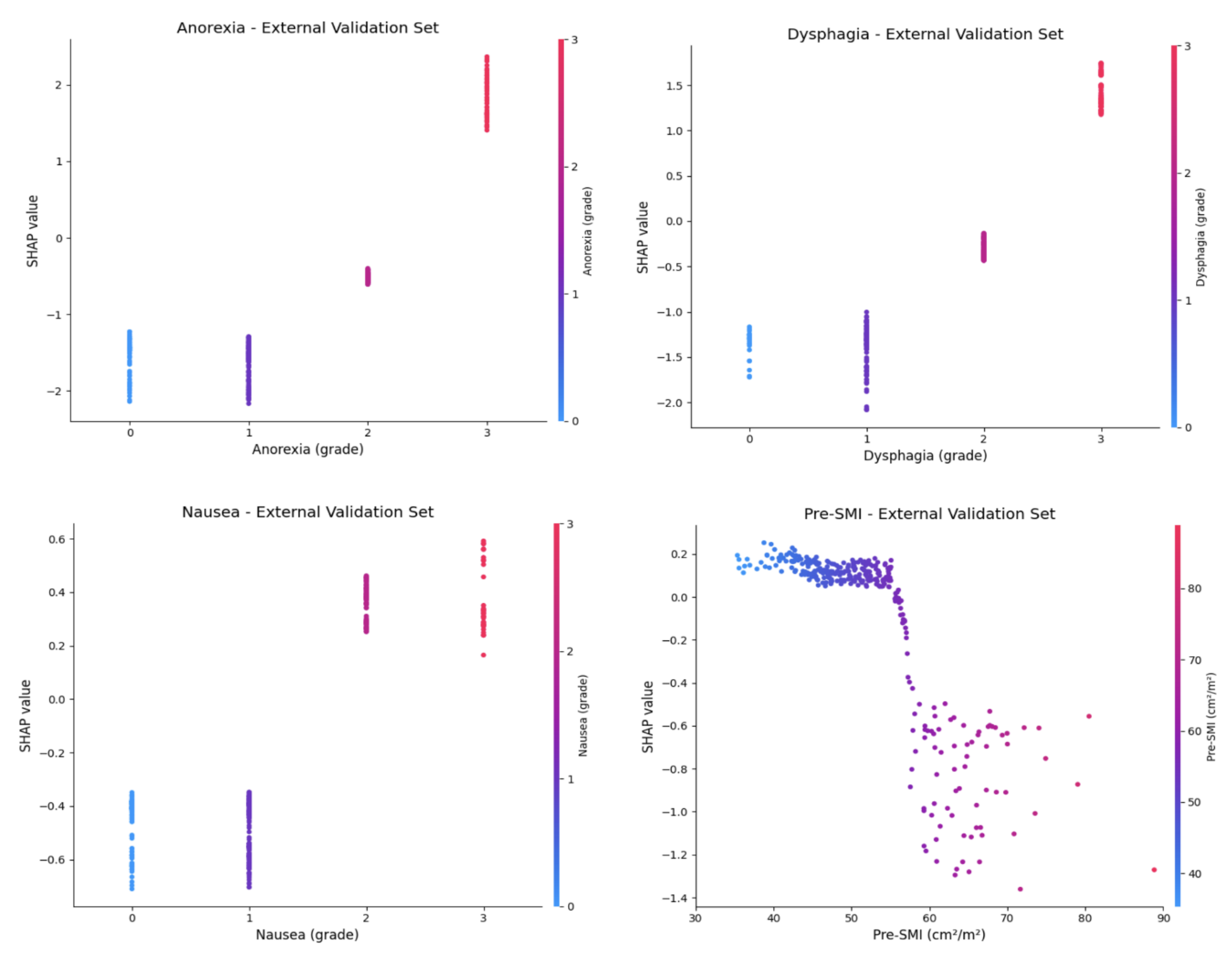
Calibration Curve



Decision Curve Analysis



SHAP Summary Plot of Catboost



SHAP Dependence Plot of Catboost

Conclusions

- **CatBoost Model Performance:** CatBoost achieved AUCs of 0.957 and 0.990 in internal and external validation, respectively, with favorable calibration and clinical net benefit.
- **Key Predictors:** Anorexia, dysphagia, and nausea were the leading SHAP predictors, followed by pre-treatment SMI and pain.
- **Clinical Implication:** Treatment-related symptoms were associated with muscle loss risk and may support on-treatment risk assessment in oral cancer patients.

References:

Suzuki, H., Asakawa, A., Amitani, H., Nakamura, N., & Inui, A. (2013). Cancer cachexia—pathophysiology and management. *Journal of Gastroenterology, 48*(5), 574–594. https://doi.org/10.1007/s00535-013-0787-0 Lee, J., Lin, J.-B., Lin, W.-C., Jan, Y.-T., Leu, Y.-S., Chen, Y.-J., & Wu, K.-P. (2025). Identifying threshold of CT-defined muscle loss after radiotherapy for survival in oral cavity cancer using machine learning. *European Radiology, 35*(7), 4289–4299. https://doi.org/10.1007/s00330-024-11303-4